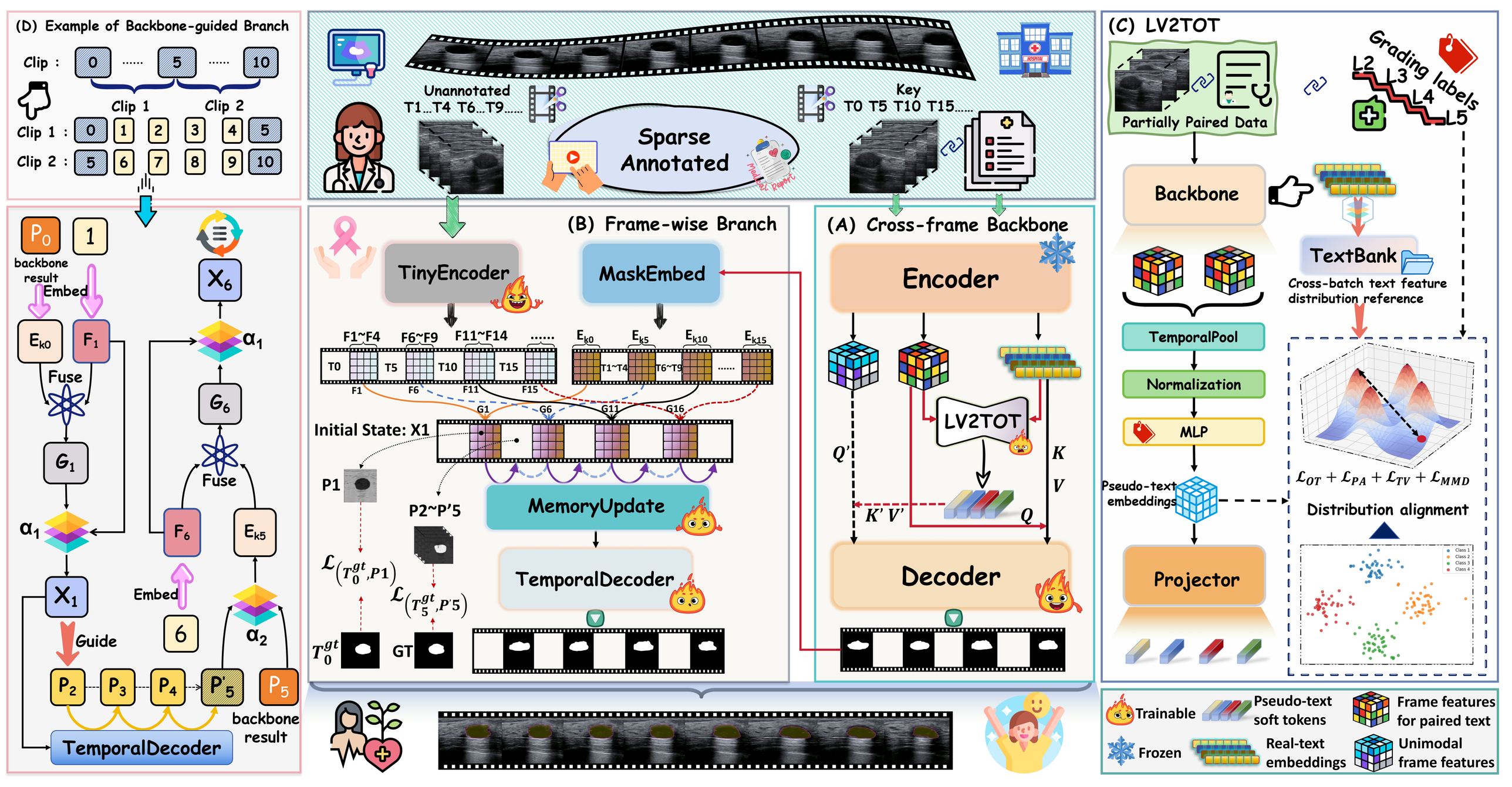

(D) Example of Backbone-guided Branch
Clip :
Clip 1
Clip 2
Clip 1 :
Clip 2 :
backbone result
Embed
Fuse
Guide
Embed
Fuse
backbone result
TemporalDecoder
Unannotated
T1...T4 T6...T9.....
Sparse Annotated
Key
T0 T5 T10 T15.....
(B) Frame-wise Branch
TinyEncoder
MaskEmbed
F1~F4
F6~F9
F11~F14
Initial State: X1
MemoryUpdate
TemporalDecoder
P1
P2~P'5
GT
(A) Cross-frame Backbone
Encoder
LV2TOT
Decoder
(C) LV2TOT
Partially Paired Data
Grading labels
Backbone
TextBank
Cross-batch text feature distribution reference
TemporalPool
Normalization
MLP
Pseudo-text embeddings
Projector
Distribution alignment
Trainable
Frozen
Pseudo-text soft tokens
Real-text embeddings
Frame features for paired text
Unimodal frame features